Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Statement of the main problem

- We are now interested in solving teh following coupled ode system

$$\begin{cases} i\frac{\partial}{\partial t}\psi(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\psi(x,t), \\ i\frac{\partial}{\partial t}\phi_\omega(x,t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t)\right)\phi_\omega(x,t) + e^{i\omega t}\hat{p}\,\psi(x,t), \end{cases} \tag{1}$$

- So we want our drivers to solve the following general case as efficiently as possible

$$i\partial_t\phi_\omega(x,t) = H(t)\phi_\omega(x,t) + S(t), \qquad H(t) = H_0 + F(t)x, \tag{2}$$

- Considering, the solution to the homogeneous problem, without the source

$$i\partial_t U(t,t_0) = \left(H_0 + F(t)x\right)U(t,t_0), \qquad U(t,t_0) = \mathbb{1}, \tag{3}$$

  we can them, from Duhamel principle write

$$\partial_t\phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,t) - i\int_{t_0}^{t_1} U(t-\tau)S(\tau)d\tau. \tag{4}$$

  The above is the exact formulation of our problem.

Geometric operators

Norm-preserving propagation schemes

$$\partial_t \phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,t) - i\int_{t_0}^{t_1} U(t-\tau)S(\tau)d\tau. \tag{4}$$

- Our test system will be a forced QHO, with source

$$\begin{cases} i\partial_t \psi(x,t) & = \left(-(1/2)\partial_x^2 + (1/2)x^2 + F(t)x\right)\psi(x,t), \qquad (\hbar = \omega = m = 1) \\ i\partial_t \phi_\omega(x,t) & = \left(-(1/2)\partial_x^2 + (1/2)x^2 + F(t)x\right)\phi_\omega(x,t) - ie^{i\omega t}\partial_x \psi(x,t), \end{cases} \tag{5}$$

- The evaluation of both the terms shall be done with matching order methods
- We will try the following geometric (norm-preserving) propagators
  - Midpoint Duhamel (order 2)
  - Yoshida scheme (order 4)
  - Runge-Kutta 4 (EFRK4)?
  - etc...

Convergence criteria

$$\partial_t y = Ay + S(x,t). \tag{6}$$

- Assessment of these schemes should be done based on two different tests with the number of steps N, and the time step h are such as

$$N_\ell = N_0 2^\ell, \qquad h_\ell = \frac{T}{N_0 2^\ell}. \tag{7}$$

  - Absolute convergence

$$p_\ell^{\mathrm{abs}} = \frac{\log(E_{\ell-1}^{\mathrm{abs}}/E_\ell^{\mathrm{abs}})}{\log(2)}, \qquad \text{with} \qquad E_\ell^{\mathrm{abs}} = \left| y_\ell(T) - y_{\mathrm{exact}}(T) \right|. \tag{8}$$

  - Self-consistency

$$p_\ell^{\mathrm{self}} = \frac{\log\left(D_{\ell-1}/D_\ell\right)}{\log(2)} \qquad \text{with} \qquad D_\ell = \left| y_\ell(T) - y_{\ell-1}(T) \right|, \qquad D_\ell \sim C h_\ell^p. \tag{9}$$

- However if we have to fix the test case to the inhomogeneous Schrödinger equation, then we can realistically only carry the second one here

Geometric operators

Yoshida (1990) - 4th-order symmetric decomposition

$$\partial_t \phi_\omega(x,t) = U(t,t_0)\phi_\omega(x,t) - i\int_{t_0}^{t_1} U(t-\tau)S(\tau)d\tau. \tag{4}$$

- From the following Strang operator $U(t) = e^{-iE_n h/2} e^{-iV_{\text{ext}}(t)h} e^{-iE_n h/2}$, we compose

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \qquad \text{with} \qquad c_1 = \frac{1}{2-2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2-2^{1/3}}. \tag{10}$$

- More precisely, we can observe the following

  Theorem 8.6 (Hairer, Solving ODE II p.554)
  If Ψ_h represent a one-step method that is symmetric, of order $p = 2k$ and defined in the domain, then the relations

$$2c_1 + c_2 = 1, \qquad 2c_1^{p+1} + c_2^{p+1} = 0, \tag{11}$$

  imply that the composition (10) is of order $p+2$.

  If you start from what is commonly known as the split-operator method, then in the above $k = 1$.

Geometric operators

Yoshida (1990) - Results 1 - Self-consistency test

$$\phi^{\omega}_{n+1} = U(\Delta t)\phi^{\omega}_{n} - i\Delta t\, U\left(\Delta t/2\right) S_{n+1/2}. \tag{??}$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \qquad \text{with} \qquad c_1 = \frac{1}{2-2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2-2^{1/3}}. \tag{10}$$

t	$\Re\,\psi$	$\Im\,\psi$	$\|\psi\|$	$\Re\,\phi$	$\Im\,\phi$	$\|\phi\|$
-1.0000	1.000000	0.000000	1.000000	0.0573947	0.0000000	0.624813
-0.5000	0.937711	-0.226288	1.000000	0.119266	-0.0066368	0.781378
-0.3750	0.900255	-0.270304	1.000000	0.142128	-0.0109301	0.822875
-0.1250	0.803565	-0.330083	1.000000	0.184650	-0.0191649	0.877411
0.0000	0.748376	-0.343661	1.000000	0.203625	-0.0191819	0.882245
0.1250	0.692051	-0.346136	1.000000	0.222276	-0.0147149	0.876229
0.3750	0.586884	-0.322034	1.000000	0.265508	0.0066504	0.842303
0.5000	0.542795	-0.299476	1.000000	0.292510	0.0197556	0.825727
1.0000	0.447387	-0.205441	1.000000	0.422166	0.0346930	0.834686

Level	$\Delta t_0 = 0.1$ (relative)	E_ψ	E_ϕ
1	Δt_0	4.22×10^{-6}	1.45×10^{-3}
2	$\Delta t_0/2$	2.44×10^{-7}	7.93×10^{-4}
3	$\Delta t_0/4$	1.50×10^{-8}	1.97×10^{-4}
Observed order		$p_\psi = 4.03$	$p_\phi = 2.01$

$$\mathrm{Ne} = 2$$
$$\mathrm{Np} = 6$$
$$-\mathrm{xmin} = \mathrm{xrange}/2 = 7$$
$$-t_0 = \mathrm{duration}/2 = 1$$

Initial conditions

$$\psi_0(x) = (\pi)^{-1/4} e^{(-(x-x_c)^2/2)}$$
$$\phi^{\omega}_0(x) = \partial_x \psi_0(x)$$
$$x_c = 0$$

- Results of the time propagation for the coupled forced QHO with and without source (Eq.5)

$$p^{\mathrm{self}}_k = \left(\mathrm{err}_{k-1}/\mathrm{err}_k\right)/\log(2)$$

Geometric operators

Yoshida (1990) - Results 2 - Convergence against a reference solution

$$\phi^{\omega}_{n+1} = U(\Delta t)\phi^{\omega}_{n} - i\Delta t\, U\left(\Delta t/2\right) S_{n+1/2}. \quad \textbf{(??)}$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \qquad \text{with} \qquad c_1 = \frac{1}{2-2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2-2^{1/3}}. \quad (10)$$

t	$\Re\,\psi$	$\Im\,\psi$	$\Vert\psi\Vert$	$\Re\,\phi$	$\Im\,\phi$	$\Vert\phi\Vert$
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Δt	E_ψ	E_ϕ	p_ψ	p_ϕ
1.0×10^{-1}	7.79×10^{-7}	2.28×10^{-4}	–	–
5.0×10^{-2}	4.85×10^{-8}	2.82×10^{-5}	4.01	3.01
2.5×10^{-2}	3.01×10^{-9}	3.44×10^{-6}	4.01	3.04
Reference ($\Delta t = \Delta t_0/2^4$)	–	–	–	–

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$$p^{\mathrm{self}}_k = \left(\mathrm{err}_{k-1}/\mathrm{err}_k\right)/\log(2)$$